Ch 9.3-4: Support Vector Machine

Lecture 29 - CMSE 381

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Fri, Nov 17, 2023

Announcements

Last time:

- 9.2 Support Vector Classifier

This lecture:

- 9.3 Support Vector Machine

Announcements:

- HW #7 due Monday

Lec #	Date			Reading	Homeworks
20	Fri	Oct 27	Dimension Reduction	6.3	
21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
23	Fri	Nov 3	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
24	Mon	Nov 6	Regression Splines	7.4	~~HW #6 Due~~
25	Wed	Nov 8	Decision Trees	8.1	HW #6 Due
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Section 1

Last Time

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

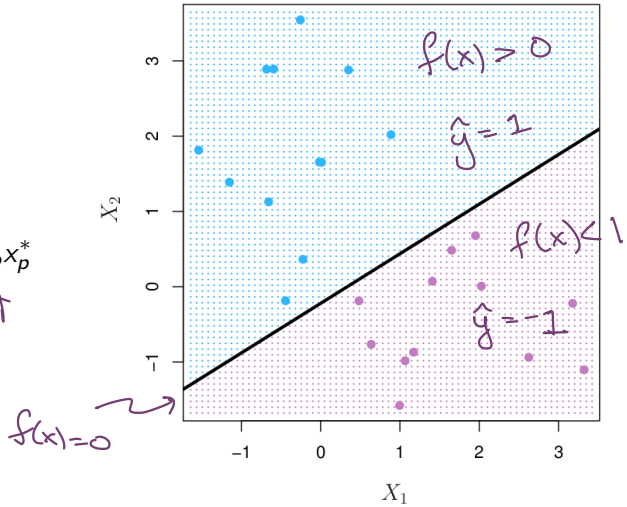
Hyperplane becomes a classifier

If you have a separating hyperplane:

- Check

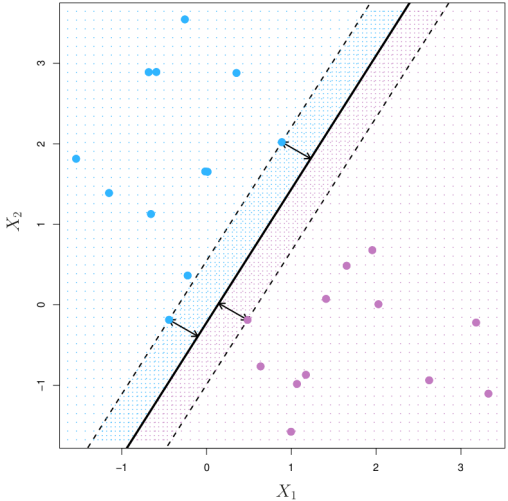
$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



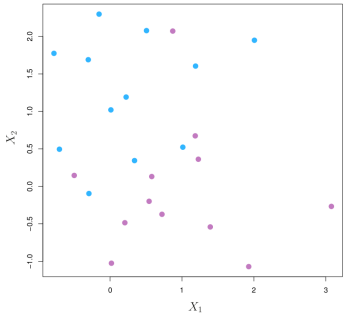
How do we pick? Old version

Maximal margin classifier

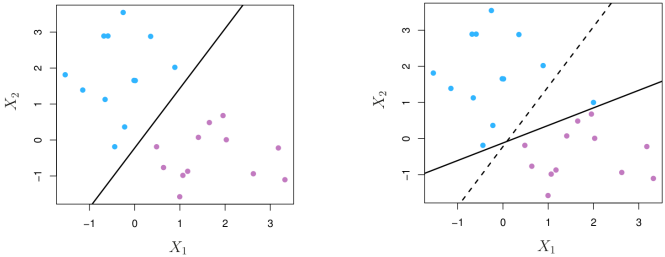


- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

Issues

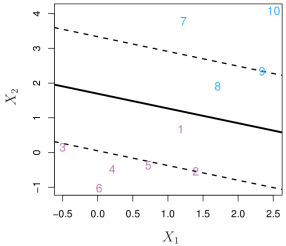


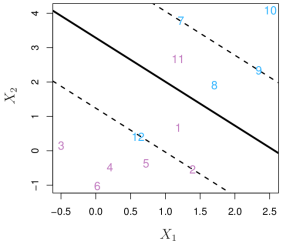
No separating hyperplane exists



Choice of hyperplane is sensitive to new points

Support Vector Classifier





Soft margin

Support vectors
- points on wrong side of margin
- points on wrong side of hyperplane

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

hyperparam.

Two formulations side by side

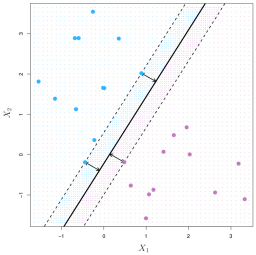
Non

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$



Wed

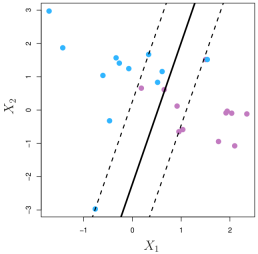
Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$



So many variables

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

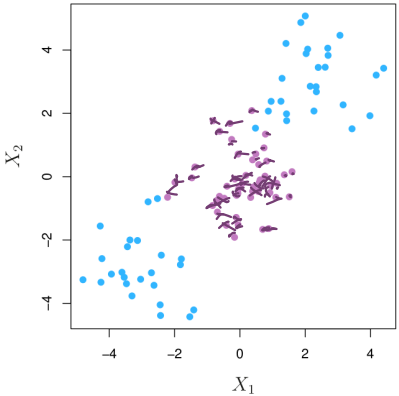
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

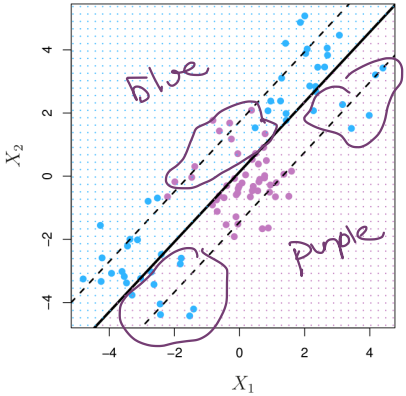
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- C is nonnegative tuning parameter
- M is the width of the margin
- $\varepsilon_1, \cdots, \varepsilon_n$ are slack variables allowing observations to go to the other side

Limiting factor of SVC





* restricted to linear boundary between the classes

Section 2

Support Vector Machine

Example of using more features

Want $2p$ features:

$$X_1, X_1^2, X_2, X_2^2, \cdots, X_p, X_p^2$$

Optimization becomes:

$$\underset{\beta_0,\beta_{11},\beta_{12},\ldots,\beta_{p1},\beta_{p2},\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to } y_i\left(\beta_0 + \sum_{j=1}^{p}\beta_{j1}x_{ij} + \sum_{j=1}^{p}\beta_{j2}x_{ij}^2\right) \geq M(1-\epsilon_i),$$

$$\sum_{i=1}^{n}\epsilon_i \leq C, \;\; \epsilon_i \geq 0, \;\; \sum_{j=1}^{p}\sum_{k=1}^{2}\beta_{jk}^2 = 1.$$

This is fine

but

Computationally bad

... why not start increasing power, or using all combos,

... we don't want to do this.

Kernels

Main idea: enlarge the feature space
(Similar to above)

But ... make it more computationally efficient

Inner products

$$\langle a, b \rangle = \sum_{i=1}^{r} a_i b_i$$

$(a_1, a_2, \ldots, a_r)$

$(b_1, b_2, \ldots, b_r)$

$a_1 b_1 + a_2 b_2 + a_3 b_3 + \ldots + a_r b_r$

Quick computations

What are the following?

- $\langle (1,1),(0,3) \rangle$ $= 1 \cdot 0 + 1 \cdot 3 = 3$
- $\langle (1,1),(3,2) \rangle$ $= 1 \cdot 3 + 1 \cdot 2 = 5$
- $\langle (2,3),(0,3) \rangle$ $= 2 \cdot 0 + 3 \cdot 3 = 9$
- $\langle (2,3),(3,2) \rangle$ $= 2 \cdot 3 + 3 \cdot 2 = 12$

SVC via inner products

$f(\underline{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$

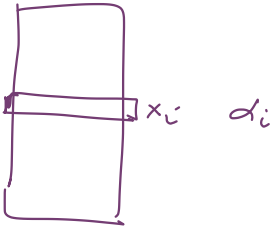
hyperplane $f(x) = 0$

$$f(\underline{x}) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle$$

x_i = the ith training point

Via magic

there are α_i for which the SVC can be written in this form

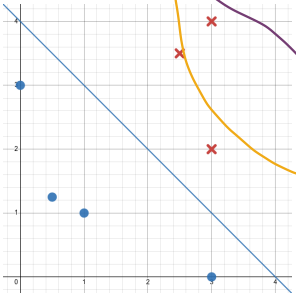


Example

$f(x_1, x_2) =$

$$-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_2 = 0$$

$$-2\sqrt{2} + \frac{\sqrt{2}}{18}\langle (X_1, X_2), (0,3)\rangle + \frac{\sqrt{2}}{6}\langle (X_1, X_2), (3,2)\rangle = 0$$



$$f(1,1) = -2\sqrt{2} + \frac{\sqrt{2}}{2}(1) + \frac{\sqrt{2}}{2}(1) = -\sqrt{2}$$

$$f(1,1) = -2\sqrt{2} + \frac{\sqrt{2}}{18}(3) + \frac{\sqrt{2}}{6}(5)$$

$$= \left(-2 + \frac{3}{18} + \frac{5}{6}\right)\sqrt{2} = -\sqrt{2}$$

What are the α_is?

Data point	α_i
$(3,4)$	0
$(2.5,3.5)$	0
$(3,2)$	$\frac{\sqrt{2}}{6}$
$(3,0)$	0
$(0,3)$	$\frac{\sqrt{2}}{18}$
$(1,1)$	0
$(0.5,1.25)$	0

What α_i's are needed to write the hyperplane

$$-2\sqrt{2} + \underbrace{\frac{\sqrt{2}}{18}\langle (X_1,X_2),(0,3)\rangle} + \underbrace{\frac{\sqrt{2}}{6}\langle (X_1,X_2),(3,2)\rangle}$$

of the previous page in the form

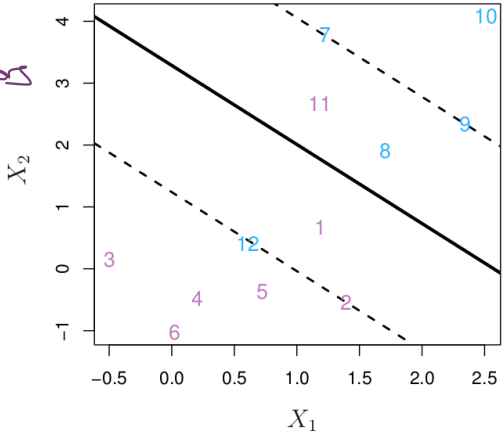
$$f(x) = \beta_0 + \sum_{i\in\mathcal{S}} \alpha_i \langle x, x_i\rangle ?$$

$\langle (x_1,x_2), (3,4) \rangle$

SVC via inner products of support vectors

The non-zero α_i's are matched with the support vectors x_i

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \underbrace{\langle x, x_i \rangle}_{\text{Change this!}}$$



The kernel

$$K(x_i, x_i')$$

* Takes in two input data points and returns some number
* $K(x_i, x_i') = \sum_{j=1}^{p} x_{ij} \cdot x'_{ij}$

  is called a <u>linear</u> <u>kernel</u>

  $\langle x_i, x_i' \rangle$

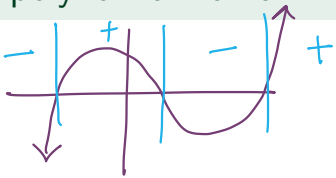
$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

* the function learned depends on the choice of kernel
* $K(x, x')$ can be thought of as a similarity function
* Learning functions like above with non-linear kernel = Support Vector Machine

A polynomial kernel

$f(x) = \beta_0 + \alpha_1 K(x, x_1) + \alpha_2 (K(x, x_2))$

$+ \cdots$



$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^{d}$$

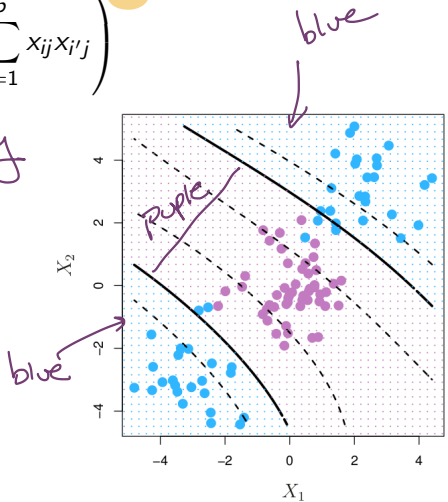
* more flexible decision boundary
* This example is a deg 3 polynomial kernel $d=3$

$\boxed{d=2}$

$K((x_1, x_2), (1, 3))$

$= (1 + x_1 \cdot 1 + x_2 \cdot 3)^2$

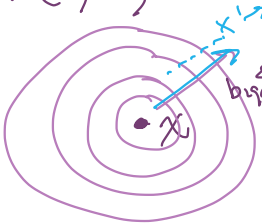
$= (1 + x_1 + 3x_2)^2$



A radial kernel

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right) = e^{-\gamma \sum (x_{ij} - x_{ij}')^2}$$

$K(x, x')$



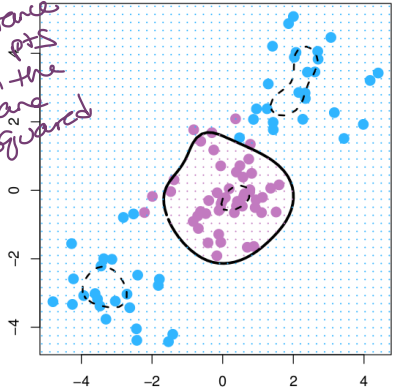
$\sum (x_j - x_j')^2$ ← bigger

← distance b/t pts in the plane squared

$e^{-\gamma \cdot \text{big}}$

$\Longrightarrow$ approaches 0

The contribution of x_i is small for x far away.



Support Vector Machine

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

Pick Kernel in advance

Coding

Section 3

SVM with more than two classes

One-Vs-One Classification

Also called all-pairs

not the kernel

* Classification problem with $K > 2$ classes

* Build $\binom{K}{2}$ SVMs where each compares a pair of classes

Ex) $y \in \{\text{apples, oranges, bananas}\}$

1st classifier apples vs oranges

2nd apples vs bananas

3rd oranges vs bananas

New data point: predict the class most commonly predicted among these

One-Vs-All Classification

Have $K > 2$ classes to predict

Fit K SVMs, each one comparing $\left\{ \begin{array}{l} \text{in the class} \\ \text{-vs- not in the class} \end{array} \right\}$

1st SVM: apples vs not apples $\rightarrow f_1(x) = \beta_0 + \beta_{1k} x_1 + \dots + \beta_{pk} x_p$

2nd SVM: bananas vs not banana $\rightarrow f_2(x)$

3rd SVM: oranges vs not oranges $\rightarrow f_3(x)$

$\hookrightarrow$ predict observation k for which $f_k(x)$ is the largest

TL;DR

SVM

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \underbrace{K(x, x_i)}_{\text{Kernel}}$$

Kernels

- Linear

$$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij} x_{i'j}$$

- Polynomial

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

- Radial

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

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